

K-Means Clustering Practical File

Aim

To implement K-Means Clustering on a given dataset using Python and group the points into clusters based on Euclidean distance.

Objectives

- Understand the concept of unsupervised learning.
- Implement K-Means clustering algorithm using Python.
- Calculate Euclidean distances between points and centroids.
- Update centroids after cluster formation.
- Visualize clustered data using Matplotlib.

Theory

K-Means Clustering is an unsupervised machine learning algorithm used to divide data into K clusters. The algorithm works by selecting initial centroids and assigning each point to the nearest centroid. After cluster formation, new centroids are calculated by taking the mean of all points in the cluster. This process repeats until the centroids stop changing significantly.

Algorithm Steps

1. Select the number of clusters K.
2. Choose initial centroids.
3. Calculate Euclidean distance of each point from centroids.
4. Assign points to the nearest cluster.
5. Compute new centroids using mean values.
6. Repeat until convergence.

Formula Used

Euclidean Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Dataset Used

P1=[0.1,0.6]
P2=[0.15,0.71]
P3=[0.08,0.9]
P4=[0.16,0.85]

P5=[0.2,0.3]
P6=[0.25,0.5]
P7=[0.24,0.1]
P8=[0.3,0.2]

Expected Output

Cluster C1 = P1, P2, P3, P4, P6
Cluster C2 = P5, P7, P8

Updated m1 = (0.148, 0.712)
Updated m2 = (0.247, 0.2)

Code Explanation

- NumPy is used for mathematical calculations.
- Matplotlib is used for visualization.
- Euclidean distance is calculated using `np.linalg.norm()`.
- Points are assigned to the nearest centroid.
- New centroids are calculated using `mean()`.
- Scatter plot visualizes the clusters and centroids.

Instructions to Run .py File in Anaconda

1. Open Anaconda Prompt.
2. Install required libraries using:
`pip install numpy matplotlib`
3. Save the Python file as `kmeans.py`
4. Navigate to the file location using `cd` command.
5. Run the program using:
`python kmeans.py`

Instructions to Run .ipynb File in Jupyter Notebook

1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open the .ipynb notebook file.
4. Click on 'Run All Cells'.
5. Observe output and visualization graph.

Advantages of K-Means

- Simple and easy to implement.

- Efficient for large datasets.
- Fast convergence.
- Useful in pattern recognition and data analysis.

Applications

- Customer segmentation
- Image segmentation
- Recommendation systems
- Market analysis
- Medical data analysis

Outcome

The K-Means clustering algorithm was successfully implemented. The given points were grouped into two clusters, and updated centroids were calculated correctly. Visualization helped in understanding cluster formation.

Conclusion

The experiment demonstrated the working of the K-Means clustering algorithm using Euclidean distance. The clusters were formed successfully and centroid values were updated accurately.

Important Viva Questions with Answers

Question	Answer
What is K-Means?	An unsupervised learning algorithm used for clustering.
What is a centroid?	The mean position of all points in a cluster.
Which distance formula is used?	Euclidean Distance.
What is unsupervised learning?	Learning without labeled data.
What is K in K-Means?	Number of clusters.
Why is K-Means called iterative?	Because centroid updating repeats until convergence.
Which libraries are used?	NumPy and Matplotlib.
What is convergence?	When centroid values stop changing significantly.

Prepared for Practical Examination